BMC Psychology

RESEARCH

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Associations among problematic smartphone use, depression, and sleep disturbance in Chinese college students: a three-wave longitudinal study

Lujie Zhong[1], Guoyi Zhu[2], Jiaying Li[1], Xueqin Zhang[1], Dongfang Wang[3*] and Yun Li[1*]

Abstract

Introduction Sleep disturbance (SD) is highly prevalent among college students and has been consistently linked to depressive symptoms (DEP). Problematic smartphone use (PSU), increasingly common among young people, has similarly been implicated in sleep problems and psychological distress. Nevertheless, the longitudinal relationship between PSU and SD and the role of DEP within this association remain insufficiently understood.

Methods Using three-wave longitudinal data collected between April 2024 and April 2025 from 6,153 Chinese college students, this study applied a random intercept cross-lagged panel model (RI-CLPM) and cross-lagged panel network analysis (CLPN) to examine the bidirectional associations among PSU, DEP, and SD and identify core and bridge symptom nodes.

Results At the between-person level, PSU, DEP, and SD were significantly and positively correlated. At the within-person level, increases in depressive symptoms consistently predicted subsequent increases in PSU and SD, whereas no stable bidirectional relationship was found between PSU and SD. CLPN analysis identified "Inattention," "Withdrawal," "Tolerance/Compulsive use," "Daily life affected," and "Early awakening" as core nodes with high expected influence, while "Concentration" emerged as a key bridge node across symptom domains.

Conclusions These findings suggest that PSU and SD may reflect behavioral and functional impairments associated with depressive symptoms rather than a stable reciprocal relationship. Depressive symptoms may be an important underlying factor in the co-occurrence of PSU and SD. These findings may provide useful implications for early screening and interventions targeting depressive symptoms, attentional/concentration difficulties, and addiction-related PSU symptoms to reduce the co-occurrence of PSU and SD and promote psychological adjustment and sleep health among college students.

Keywords Problematic smartphone use, Depression, Sleep disturbance, College students, Longitudinal study

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Introduction

Sleep disturbance (SD) is characterized by impaired sleep quality and/or abnormal sleep duration [1]. This issue has become a major public health concern worldwide, affecting an estimated 30%–45% of the global population [2]. Among college students, SD is particularly prevalent, likely due to multiple factors such as academic stress and challenges related to social adaptation [3]. A large-scale meta-analysis of Chinese college students reported that approximately 25.7% experienced SD, and about one in five students were dissatisfied with their sleep quality [4]. In addition, SD is closely associated with individuals' psychological health, particularly depressive symptoms [5]. Given the high prevalence of SD among young people and its severe adverse consequences, a comprehensive understanding of its risk factors and underlying mechanisms is crucial for the development of effective prevention and intervention strategies.

Problematic smartphone use (PSU) is generally defined as an excessive and poorly controlled pattern of smartphone engagement that leads to functional impairment and psychological distress in daily life [6], and has been documented with considerable prevalence among young people [7]. Empirical evidence suggests that PSU adversely affects individuals' sleep-related functioning [8], likely through multiple pathways. At a behavioral level, excessive smartphone use may impair sleep quality by delaying bedtime, increasing nighttime awakenings, and shortening sleep duration [9, 10]. Beyond these direct effects, PSU is often associated with higher levels of anxiety, depression, and psychological stress, each of which constitutes an established risk factor for sleep disturbance [11, 12].

However, findings regarding the relationship between PSU and SD remain inconsistent. While one study reported only a unidirectional predictive effect of PSU on SD [13], others suggested a bidirectional relationship between the two [14, 15]. In addition, previous research has proposed that depression (DEP) may mediate the association between PSU and SD [8]; yet the specific pathways and underlying dynamic processes remain poorly characterized. To address these limitations, the present study employed three-wave longitudinal data and applied the random intercept cross-lagged panel model (RI-CLPM) and cross-lagged panel network analysis (CLPN) to examine the bidirectional relationship between PSU and SD, as well as the mediating role of DEP.

The relationship between PSU and SD

PSU has been theoretically and empirically linked to subsequent SD through several distinct mechanisms. From the sleep displacement perspective, greater smartphone engagement reduces the time available for sleep, contributing to delayed sleep onset, shortened sleep duration, and poorer sleep quality [16]. In addition, frequent smartphone use before bedtime or during the night may interfere with sleep by exposing individuals to screen-emitted blue light and emotionally engaging content, thereby elevating physiological and cognitive arousal and disrupting the normal sleep process [17, 18]. Therefore, individuals with higher levels of PSU are more likely to experience disruptions in sleep rhythms and subsequently exhibit higher levels of SD [15, 19]. Cross-sectional studies have also shown that higher levels of PSU are significantly associated with poorer sleep quality [20, 21]. A longitudinal study among Chinese college students found that problematic mobile phone use could unidirectionally predict subsequent declines in sleep quality, rather than the reverse relationship [13]. Another study similarly found that higher levels of PSU at earlier time points predicted more severe subsequent SD [15].

SD, in turn, may also predict PSU. According to the Interaction of Person-Affect-Cognition-Execution (I-PACE), individuals' affective states and cognitive responses, and executive control functions influence the development and maintenance of internet-related problematic behaviors [22]. Sleep problems are often accompanied by heightened negative affect and diminished self-regulatory capacity, which may drive individuals to use smartphones as a coping strategy—whether for emotional compensation, mood relief, or even as a sleep aid. However, such bedtime smartphone use is associated with poorer sleep quality and delayed bedtimes [16], potentially reinforcing a cycle in which SD drives further smartphone engagement and ultimately elevates the risk of PSU [23, 24]. Longitudinal evidence has further shown that insomnia symptoms positively predict subsequent levels of PSU [25].

Although previous cross-sectional studies have consistently documented a positive association between PSU and SD, they were inherently unable to reveal the directionality of this relationship. Moreover, the limited longitudinal evidence available has yielded inconsistent findings, leaving the dynamic longitudinal relationship between PSU and SD insufficiently understood.

The potential role of depressive symptoms

DEP is a common and clinically significant mental health disorder, typically involving low mood, anhedonia, disturbances to sleep and energy levels, and impaired cognitive function [26]. Depressive symptoms are particularly prevalent among college students due to factors such as academic pressure and social adjustment [27–29]. A recent systematic review reported an average worldwide prevalence rate of depressive symptoms among college students of approximately 27% [30]; in China, data

indicate that the overall prevalence of depression among college students is about 28.4% [31].

Notably, depression may play an important role in the longitudinal relationship between PSU and SD. According to the I-PACE model [22], individuals' affective and cognitive responses, as well as executive control functions, may influence the development and maintenance of internet-related problematic behaviors. Individuals with depression are often characterized by elevated levels of negative affect, rumination, and difficulties in emotion regulation, which may increase their tendency to use smartphones to alleviate negative emotions, thereby further increasing the risk of PSU [22, 32]. On the other hand, depression is also an important risk factor for sleep problems. According to Beck's cognitive theory [33], individuals with depression are more likely to exhibit negative cognitive processing, rumination, and elevated levels of emotional arousal, all of which may further disrupt normal sleep processes and increase the risk of SD [34, 35]. Previous studies have also shown that depression is closely associated with PSU [36, 37], and individuals with depression are more likely to experience SD [38]. Therefore, DEP may serve as an important psychological mechanism linking PSU and SD. However, its potential mediating role in the relationship between the two variables still requires further investigation.

Current study

Although previous longitudinal studies have provided important insights into the temporal relationships among PSU, DEP, and SD, several methodological limitations remain. For example, cross-sectional designs are unable to establish the direction of causality, and the traditional cross-lagged panel model (CLPM) cannot adequately distinguish within-person effects from between-person effects. In addition, previous studies have primarily examined the relationships among PSU, DEP, and SD at the construct level, making it difficult to capture the longitudinal dynamics among individual symptoms. To address these limitations, the present study employed both the RI-CLPM and CLPN to examine the dynamic relationships among multiple variables over time and their complex interrelationships [39, 40].

While controlling for autoregressive effects, the RI-CLPM disentangles within-person from between-person variance. Its cross-lagged effects represent the prospective influence of one variable at time T on another variable at the subsequent time point (T + 1) [41], thereby providing a more robust statistical framework for examining the directional relationships among variables. Accordingly, the present study first employed the RI-CLPM to investigate the bidirectional longitudinal associations among PSU, DEP, and SD and further examined the longitudinal mediating role of DEP in the association between PSU and SD.

Nevertheless, the RI-CLPM remains limited in characterizing the evolving network structure of multiple variables and the dynamic relationships among individual symptoms over time. According to the network theory of psychopathology, mental disorders can be conceptualized as dynamic networks of interacting symptoms rather than manifestations of a single latent construct [42]. Because PSU, DEP, and SD each comprise multiple heterogeneous symptoms, different symptoms may play distinct roles in their longitudinal development. Therefore, analyses conducted solely at the construct level may fail to reveal the specific roles of individual symptoms in the longitudinal associations among these three constructs [43]. To overcome this limitation, the present study further employed CLPN to construct cross-temporal symptom networks, thereby characterizing the longitudinal relationships among individual symptoms and identifying both bridge nodes connecting different symptom domains and central nodes within the network [44].

Unlike the RI-CLPM, which focuses on the overall longitudinal associations among constructs, the CLPN can further identify which specific symptoms may drive the longitudinal associations between constructs and reveal potential cross-domain longitudinal connections among individual symptoms. Consequently, CLPN facilitates a deeper understanding of longitudinal associations among constructs from a symptom-level perspective, thereby facilitating the identification of key symptoms with potential intervention value and informing the development of more precise symptom-targeted intervention strategies [45].

Based on the above theoretical and empirical background, the present study proposes the following hypotheses: (1) at the between-person level, PSU, DEP, and SD are expected to be significantly and positively associated with one another; (2) at the within-person level, bidirectional longitudinal relationships may exist among PSU, DEP, and SD. Furthermore, DEP may mediate the longitudinal relationship between PSU and SD. In addition, the present study further incorporates CLPN to explore the dynamic symptom structure of PSU, DEP, and SD, and to identify key symptoms and cross-domain connection patterns across different symptom domains, thereby providing further insights into the interrelationships among these constructs.

Methods

Participants

This study employed a convenience sampling method to conduct a large-scale three-wave online longitudinal survey. Data were collected at three time points: April 15–24, 2024 (T_1), October 15–29, 2024 (T_2), and April

14–27, 2025 (T_3). The web-based survey was distributed through the "Survey Star" online platform, and the Quick Response (QR) code of the questionnaire was forwarded to participants via WeChat. All participants were required to complete the questionnaire by scanning the QR code using their mobile phones. The platform automatically verified the questionnaire's completeness, and submission was permitted only after all items had been completed.

A total of 13,943 college students participated in the T_1 survey, 13,738 in the T_2 survey, and 12,261 in the T_3 survey. To improve data quality, the following exclusion criteria were applied: (i) incorrect identity information (e.g., incomplete student number); (ii) excessively short response time; (iii) inconsistent or careless responses; and (iv) self-reported diagnosis of mental illness. After screening, 12,853, 12,087, and 10,639 valid responses were retained for T_1, T_2, and T_3, respectively. Ultimately, 6,153 college students who completed all three waves of the survey were included in the subsequent analyses. The detailed participant screening procedure is presented in Fig. 1.

This study adopted an anonymous survey methodology, with participants identified using unique student numbers. Prior to participation, all participants were required to provide electronic informed consent, which explicitly stated their right to discontinue or withdraw from the survey at any time without penalty. During the survey period, a toll-free psychological hotline was established to provide assistance to participants in need. In addition, a written report summarizing the overall status of participants was provided to the participating schools. The research protocol was approved by the Ethics Committee of the School of Psychology at South China Normal University (Approval No.: SCNU-PSY-2023-345).

Measurements

Problematic smartphone use was assessed using the Smartphone Addiction Scale–Short Version (SAS-SV) [46], which measures the severity of maladaptive and excessive smartphone use behaviors. The SAS-SV consists of 10 items covering core symptoms of problematic smartphone use, including withdrawal, tolerance/compulsive use, preoccupation, and functional impairment. Participants rated each item on a six-point Likert scale (1 = strongly disagree to 6 = strongly agree), yielding a total score ranging from 10 to 60, with higher scores indicating more severe problematic smartphone use. In the present study, minor wording adaptations were made to improve cultural relevance for Chinese college students. Specifically, references to "Twitter or Facebook" in the original scale were replaced with "WeChat or QQ," which are more commonly used social networking platforms in the Chinese context. In the present study, the SAS-SV showed excellent internal consistency across the three waves (Cronbach's $\alpha = 0.948–0.957$).

Depression was assessed using the Patient Health Questionnaire-9 (PHQ-9) [47], which measures the

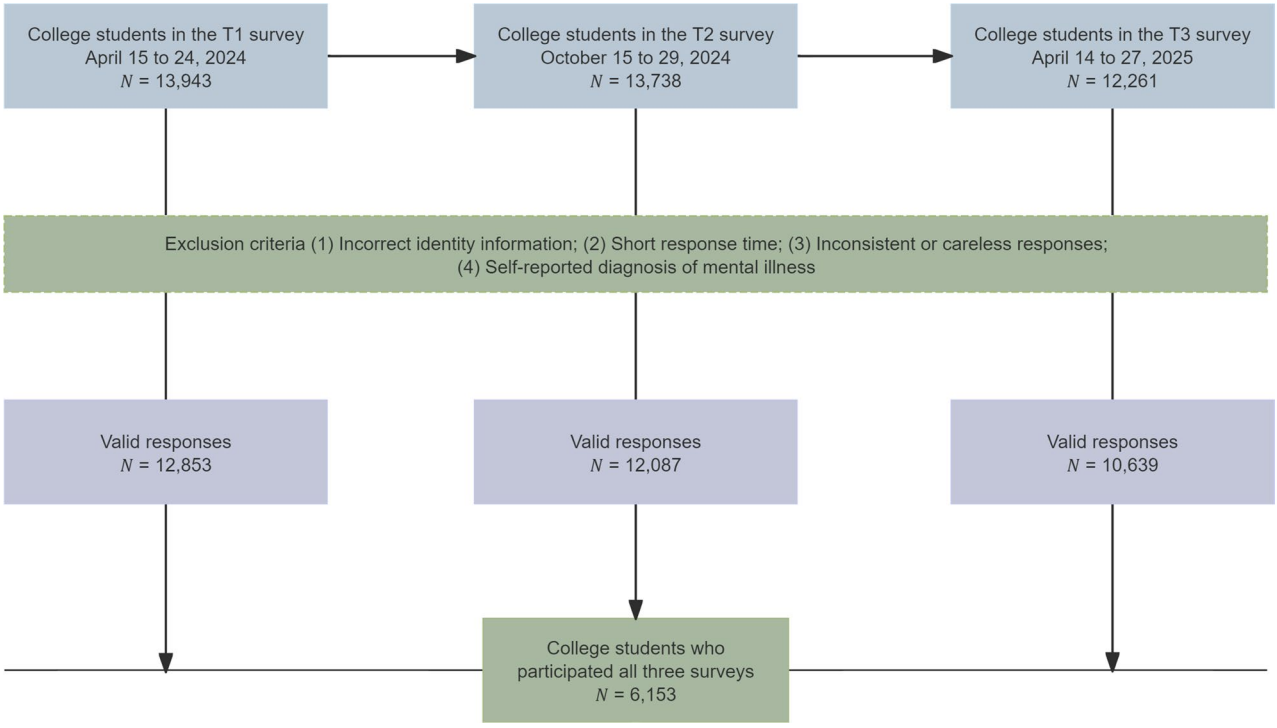


Fig. 1 Consort diagram for the number of students that participated in each wave

frequency of depressive symptoms experienced over the past two weeks. The PHQ-9 consists of nine items corresponding to the diagnostic criteria for major depressive disorder in the DSM-IV. Each item is rated on a four-point scale (0 = not at all to 3 = nearly every day), yielding a total score ranging from 0 to 27, with higher scores indicating more severe depressive symptoms. In the present study, the PHQ-9 demonstrated excellent internal consistency across the three waves, with Cronbach's α coefficients ranging from 0.937 to 0.946.

Sleep disturbance during the past month was measured using the Youth Self-Rating Insomnia Scale (YSIS) [48], which consists of four items assessing subjective sleep quality and insomnia symptoms. Participants were asked: "How would you rate the quality of your sleep overall?" (subjective sleep quality), "How often would you have trouble falling asleep?" (difficulty initiating sleep), "How often would you wake up frequently during the night?" (difficulty maintaining sleep), and "How often would you wake up very early and can't get back to sleep?" (early morning awakening). Subjective sleep quality was rated on a 5-point scale ranging from 1 (very good) to 5 (very poor). The remaining three items used the same 5-point response format: never, < 1 time/week (rarely), 1–2 times/week (sometimes), 3–5 times/week (often), and 6–7 times/week (almost every day). Higher scores indicated poorer sleep quality. Cronbach's α ranged from 0.802 to 0.819 across waves in the present study.

Covariates included age, gender (1 = male, 2 = female), residence (1 = rural, 2 = town, 3 = urban), and only-child status (1 = yes, 2 = no), all of which were measured at T_1. These variables are commonly used as covariates in research and were included to control for confounding factors and to improve the models' explanatory power. Previous studies have suggested that different age groups may differ in lifestyle, behavioral patterns, and smartphone use behaviors [49–51], and these factors are closely associated with depression and sleep problems [21]. Gender differences are also evident, with females generally reporting higher levels of depressive symptoms and sleep disturbance [52], while differences in problematic smartphone use behaviors have also been observed across genders [53]. In addition, demographic variables such as "residence" and "only-child status" may also be associated with individuals' social support, as well as college students' emotional status, sleep conditions, and smartphone use behaviors [21, 54, 55].

Statistical analysis

All data analyses, including the preliminary analyses, attrition analysis, RI-CLPM and CLPN, were conducted using Mplus 8.3 [56] and R software [57]. First, descriptive statistics and correlation analyses were conducted using R software. Attrition analysis was conducted by comparing baseline demographic characteristics and study variables between participants who were retained across all three waves and those who were not retained. Second, longitudinal measurement invariance of the scales across different time points was examined using confirmatory factor analysis (CFA) in Mplus 8.3, and the longitudinal relationships among PSU, DEP, and SD were investigated using a RI-CLPM. Finally, a CLPN was constructed in R software to explore the internal associations among PSU, DEP, and SD.

Three increasingly constrained models were tested sequentially: the configural invariance model, the metric invariance model, and the scalar invariance model. At each step, additional constraints were imposed on factor loadings and intercepts based on the preceding model. Consistent with the recommendations of Chen [58] for studies with relatively large samples, measurement invariance was evaluated based on changes in model fit indices, with $\Delta CFI < 0.01$, $\Delta RMSEA < 0.015$, and $\Delta SRMR < 0.01$ indicating acceptable invariance. Subsequently, the RI-CLPM was estimated using robust maximum likelihood estimation (MLR). Model fit was considered acceptable when $\chi^2/df \leq 5$, CFI and TLI ≥ 0.90, and RMSEA and SRMR ≤ 0.08 [59]. Following established modeling procedures [60] and previous research [59], a series of increasingly constrained models was constructed to determine the final model. Specifically, Model 1 served as the baseline unconstrained model, and subsequent models sequentially imposed equality constraints on the following parameters based on the preceding model: (1) autoregressive effects; (2) cross-lagged effects; and (3) within-person variance effects. Differences in model fit were considered negligible when $\Delta CFI \leq 0.01$ and $\Delta RMSEA \leq 0.015$ [61, 62]. To facilitate the interpretation of the magnitude of cross-lagged effects, standardized coefficients of 0.03, 0.07, and 0.12 were considered small, medium, and large effects, respectively, based on the recommendations of Orth et al. [63]. Longitudinal mediation effects were estimated using the MODEL CONSTRAINT command, and their statistical significance was evaluated based on standard errors and confidence intervals.

To further examine the dynamic internal relationships among PSU, DEP, and SD, a CLPN model was constructed. The primary aim of the CLPN analysis was to identify key nodes in the network, including nodes with relatively high outgoing expected influence, ingoing expected influence, or bridge expected influence, as these nodes may play a more critical role within the network. However, these findings were mainly intended to generate potential research hypotheses and should therefore be interpreted with caution. To improve model parsimony and reduce the risk of overfitting, LASSO regression with 10-fold cross-validation was applied to regularize

the regression coefficients, shrinking smaller coefficients toward zero. For each nodewise regression, the optimal regularization parameter (λ) was selected using the lambda.min criterion implemented in cv.glmnet based on 10-fold cross-validation [64]. Two CLPN models ($T_1 \rightarrow T_2$ and $T_2 \rightarrow T_3$) were separately estimated, with age, gender, residence, and only-child status entered as control covariates to adjust for potential confounding effects. Regression coefficients were estimated using the glmnet package, and network visualization was conducted using the qgraph package [64, 65]. In addition, to enhance the readability of the network figures, paths with absolute regression coefficients smaller than 0.07 were omitted from visualization. At the node level, centrality indices were used to evaluate the relative importance of nodes within the network. Specifically, out-expected influence reflected the overall predictive effect of a node on other nodes, whereas in-expected influence reflected the extent to which a node was influenced by other nodes. Nodes with relatively high bridge expected influence were considered to play a key role in connecting different symptom domains. The robustness of the network results was evaluated using the bootnet package [66]. To assess the accuracy of edge-weight estimation, 1,000 nonparametric bootstrap resamples were performed to generate 95% confidence intervals (CIs), with narrower intervals indicating higher estimation accuracy. In addition, the stability of edge weights and centrality indices was assessed using the case-drop bootstrap procedure, in which the network was repeatedly re-estimated after randomly removing different proportions of the sample, and correlation stability coefficients were calculated. The coefficient represents the maximum proportion of cases that can be dropped while maintaining a correlation of at least 0.7 with the original centrality estimates. Generally, a correlation stability coefficient above 0.25 indicates acceptable stability, whereas a value above 0.5 indicates good stability [66]. Finally, centrality difference tests were conducted to examine whether differences in centrality indices between nodes were statistically significant.

Results

Preliminary results

A total of 6,153 participants were included in the analysis, including 4,906 women (79.73%) and 1,247 men (20.27%). The mean age of the participants was 20.06 years ($SD = 1.36$). Regarding residence, 2,927 participants (47.57%) were from rural areas, 1,752 (28.47%) were from town areas, and 1,474 (23.96%) were from urban areas. In addition, 1,140 participants (18.53%) were only children, whereas 5,013 (81.47%) had siblings. All correlations among PSU, DEP, and SD across the three waves were significant (see Table 1). The content, abbreviations, and labels of all PSU, DEP, and SD items are presented in Supplementary Table 1. The results of the measurement invariance analysis indicated that PSU, DEP, and SD achieved acceptable longitudinal measurement invariance, as the changes in fit indices among the configural invariance, metric invariance, and scalar invariance models remained within the recommended thresholds (see Table 2).

Attrition analysis

To evaluate whether sample attrition introduced potential selection bias, baseline characteristics were compared between participants who were retained across all three survey waves (n = 6,153) and those who were not retained (n = 6,700) (see Table S2). Participants who completed all three waves were younger ($t = 19.24$, $p < 0.001$, $d = 0.34$) and included a higher proportion of females ($\chi^2 = 616.22$, $p < 0.001$, Cramér's $V = 0.22$). Statistically significant differences were also observed in baseline depressive symptoms ($t = 5.36$, $p < 0.001$, $d = 0.09$), problematic smartphone use (t = -2.90, $p = 0.004$, $d = 0.05$), residence (χ^2 = 6.73, $p = 0.035$, Cramér's $V = 0.02$), and only-child status (χ^2 = 34.72, $p < 0.001$, Cramér's $V = 0.05$); however, the corresponding effect sizes were small or negligible. No significant difference was found in sleep disturbance ($t = 1.92$, $p = 0.055$, $d = 0.03$). Overall, except for age and gender, baseline differences between retained and attrition participants were small. Given that age, gender, residence, and only-child status were included as covariates

Table 1 Descriptive statistics and pearson correlation ($N = 6,153$)

Variables	M ± SD	1	2	3	4	5	6	7	8	9
1. T1 PSU	27.84 ± 10.72	1								
2. T2 PSU	27.15 ± 10.40	0.530***	1							
3. T3 PSU	25.44 ± 10.62	0.495***	0.546***	1						
4. T1 DEP	13.73 ± 4.49	0.480***	0.370***	0.366***	1					
5. T2 DEP	13.24 ± 4.35	0.338***	0.454***	0.382***	0.527***	1				
6. T3 DEP	12.52 ± 4.15	0.302***	0.365***	0.448***	0.465***	0.560***	1			
7. T1 SD	6.83 ± 2.60	0.278***	0.207***	0.226***	0.451***	0.329***	0.287***	1		
8. T2 SD	6.68 ± 2.53	0.240***	0.279***	0.253***	0.394***	0.457***	0.353***	0.577***	1	
9. T3 SD	6.29 ± 2.42	0.253***	0.272***	0.314***	0.385***	0.403***	0.460***	0.529***	0.578***	1

***$p < 0.001$

Table 2 Longitudinal measurement equivalence of research variables

	χ^2	df	CFI	TLI	RMSEA	SRMR	ΔCFI	ΔRMSEA	ΔSRMR
Measurement invariance (longitudinal)									
PSU									
Configural Invariance	18003.332	402	0.838	0.824	0.084	0.043	—	—	—
Metric Invariance	18255.471	420	0.835	0.830	0.083	0.044	−0.003	−0.001	0.001
Scalar Invariance	18438.944	438	0.834	0.835	0.082	0.044	−0.001	−0.001	0.000
DEP									
Configural Invariance	5521.327	321	0.923	0.916	0.051	0.031	—	—	—
Metric Invariance	5545.492	337	0.923	0.920	0.050	0.033	0.000	−0.001	0.002
Scalar Invariance	5656.228	353	0.922	0.922	0.049	0.033	−0.001	−0.001	0.000
SD									
Configural Invariance	2228.758	51	0.895	0.864	0.083	0.042	—	—	—
Metric Invariance	2244.342	57	0.894	0.878	0.079	0.042	−0.001	−0.004	0.000
Scalar Invariance	2305.860	63	0.892	0.886	0.076	0.043	−0.002	−0.003	0.001

χ^2 Yuan-Bentler scaled chi-square, *df* degrees of freedom, *CFI* Comparative fit index, *TLI* Tucker–Lewis index, *RMSEA* Root mean square error of approximation, *SRMR* Standardized root mean square residual

Table 3 Model comparison test for the RI-CLPMs

Model	χ^2	df	CFI	TLI	RMSEA	SRMR	ΔCFI	ΔRMSEA	ΔSRMR
RI-CLPM 1	140.109	27	0.993	0.982	0.026	0.015	—	—	—
RI-CLPM 2	149.632	30	0.993	0.983	0.025	0.015	0.000	−0.001	0.000
RI-CLPM 3	161.865	36	0.993	0.985	0.024	0.015	0.000	−0.001	0.000
RI-CLPM 4	263.224	42	0.987	0.978	0.029	0.034	−0.006	0.005	0.019

χ^2 Yuan-Bentler scaled chi-square, *df* degrees of freedom, *CFI* Comparative fit index, *TLI* Tucker–Lewis index, *RMSEA* Root mean square error of approximation, *SRMR* Standardized root mean square residual

in the subsequent models, the potential impact of sample attrition on the longitudinal analyses is likely to be limited.

RI-CLPM results

After controlling for covariates, the RI-CLPM was used to distinguish between-person effects from within-person effects among PSU, DEP, and SD. First, four extended RI-CLPMs were constructed, among which Model 3 demonstrated the best fit (see Table 3) and was therefore selected as the final model. The final model showed a good fit to the data (χ^2 = 161.865, df = 36, CFI = 0.993, TLI = 0.985, RMSEA = 0.024, SRMR = 0.015).

The results of RI-CLPM3 (see Fig. 2) indicated significant positive correlations among the random intercepts of PSU, DEP, and SD, suggesting that individuals with higher baseline levels of PSU generally also exhibited higher baseline levels of DEP and SD. In addition, gender was significantly associated with DEP (β = −0.092, $p < 0.001$) and SD (β = 0.087, $p < 0.001$), indicating that male college students reported higher levels of DEP, whereas female college students reported higher levels of SD. Meanwhile, age was significantly negatively associated with PSU, DEP, and SD (β = −0.163, −0.128, and −0.087, respectively, all $p < 0.001$), suggesting that younger college students were more likely to exhibit PSU, DEP, and SD. Residence was also negatively associated with DEP and SD (β = −0.060, $p < 0.001$; β = −0.034, $p = 0.027$), indicating that students from urban areas reported lower levels of DEP and SD compared with those from rural areas.

At the within-person level, DEP positively predicted subsequent PSU ($T_1 \rightarrow T_2$, β = 0.064, $p < 0.01$, with a small effect size approaching the medium-effect threshold; $T_2 \rightarrow T_3$, β = 0.060, $p < 0.01$, with a small effect size approaching the medium-effect threshold) and SD ($T_1 \rightarrow T_2$, β = 0.088, $p < 0.001$, with a medium effect size; $T_2 \rightarrow T_3$, β = 0.091, $p < 0.001$, with a medium effect size). However, no significant within-person effects of PSU and SD on the other variables were found. In addition, the longitudinal mediating effect of DEP in the relationship between PSU and SD was not supported (PSU→DEP→SD: β = 0.000, 95% CI = [−0.001, 0.001]; SD→DEP→PSU: β = 0.002, 95% CI = [−0.007, 0.011]).The R^2 values for all outcome variables in the RI-CLPM are reported in Table 4.

CLPN results

The CLPN analysis was conducted to capture temporal relationships among nodes and to identify potential key nodes linking PSU, DEP, and SD. Two networks, $T_1 \rightarrow T_2$ and $T_2 \rightarrow T_3$, were constructed, with age, gender, residence, and only-child status included as covariates (see Fig. 3a and b, respectively). The edge-weight matrices of the networks are presented in Supplementary Tables 3 and 4. In each matrix, the coefficient within a given cell

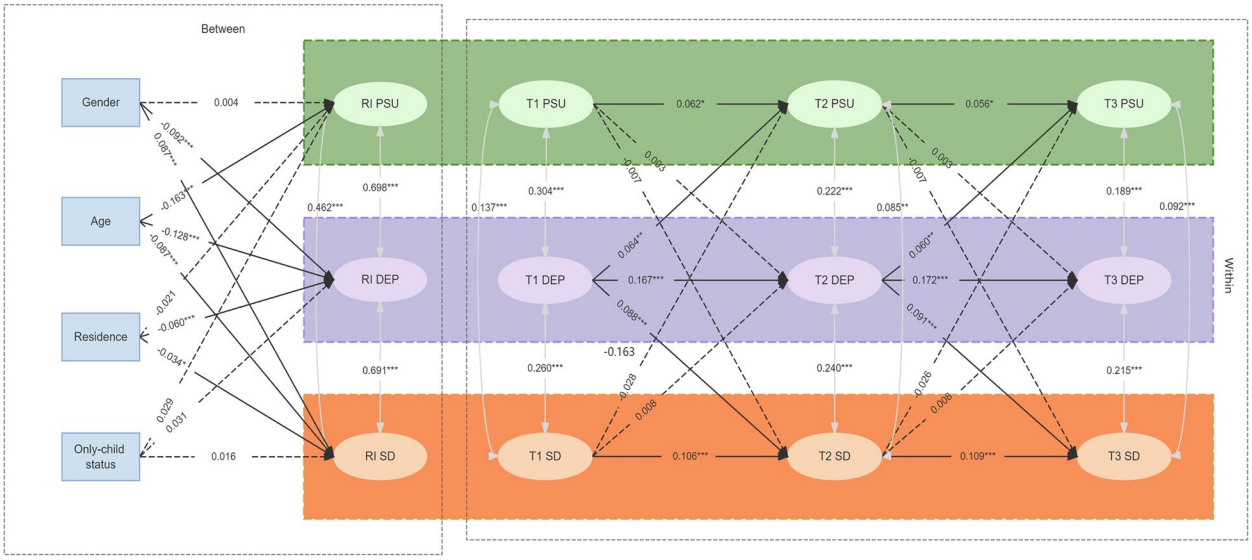


Fig. 2 Final model of RI-CLPM. Note. Solid lines represent significant paths, and dashed lines represent non-significant paths. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$. The coefficients presented in the figure are standardized

Table 4 Effect sizes (R^2) for the RI-CLPM

Model	Outcome variable	Effect size (R^2)	
		T2	T3
RI-CLPM	PSU	0.010	0.008
	DEP	0.029	0.031
	SD	0.023	0.025

represents the predictive effect of the node in the corresponding row on the node in the corresponding column. Centrality indices were subsequently estimated (see Figs. 3c and 4), and the detailed results for outgoing expected influence, ingoing expected influence, and bridge expected influence in the $T_1 \rightarrow T_2$ and $T_2 \rightarrow T_3$ networks are presented in Supplementary Tables 5 and 6, respectively. Outgoing expected influence reflects the relative influence of a node on other nodes, whereas ingoing expected influence indicates the extent to which a node is influenced by other nodes. Higher bridge expected influence suggests that a node may play a more important bridging role in linking the PSU, DEP, and SD symptom clusters.

Based on outgoing expected influence, in the $T_1 \rightarrow T_2$ network, the highest-ranking nodes in the DEP, PSU, and SD domains (labeled as "groups" in the figures for visualization purposes only) included DEP2 (Sad mood), DEP4 (Energy), and DEP7 (Concentration); PSU10 (Daily life affected), PSU4 (Withdrawal), and PSU7 (Tolerance/Compulsive use); and SD4 (Sleep quality) and SD3 (Early awakening), respectively. In the $T_2 \rightarrow T_3$ network, the highest-ranking nodes in the DEP domain were DEP1 (Anhedonia), DEP5 (Appetite), and DEP6 (Worthlessness), while PSU4 (Withdrawal) was the most prominent node in the PSU domain, with PSU10 (Daily life affected) and PSU7 (Tolerance/Compulsive use) also ranking relatively high, and SD1 (Insomnia) and SD3 (Early awakening) were the highest-ranking nodes in the SD domain. Based on ingoing expected influence, PSU2 (Inattention) consistently ranked among the relatively prominent nodes across both networks. Overall, nodes that repeatedly appeared across the two networks mainly included PSU2 (Inattention), PSU4 (Withdrawal), PSU7 (Tolerance/Compulsive use), and PSU10 (Daily life affected) in the PSU domain, as well as SD3 (Early awakening) in the SD domain. In addition, nodes with relatively high bridge expected influence included DEP2 (Sad mood), DEP4 (Energy), and DEP7 (Concentration) in the $T_1 \rightarrow T_2$ network, as well as DEP1 (Anhedonia), SD1 (Insomnia), and DEP7 (Concentration) in the $T_2 \rightarrow T_3$ network. Notably, DEP7 (Concentration) emerged as a common bridge node across both networks.

The bootstrap confidence intervals of the estimated edges for the two networks are presented in Supplementary Fig. 1. The results of the network robustness analyses based on the case-dropping procedure are shown in Supplementary Fig. 2. In the $T_1 \rightarrow T_2$ network, the correlation stability coefficients (correlation stability coefficient, CS-C) for edge weights, outgoing expected influence, ingoing expected influence, and bridge expected influence were 0.439, 0.595, 0.750, and 0.439, respectively. In the $T_2 \rightarrow T_3$ network, the corresponding CS-C values were 0.517, 0.595, 0.750, and 0.517, respectively. The results of the centrality difference tests are presented in Supplementary Figs. 3–5. Overall, both networks demonstrated acceptable to good levels of stability and robustness.

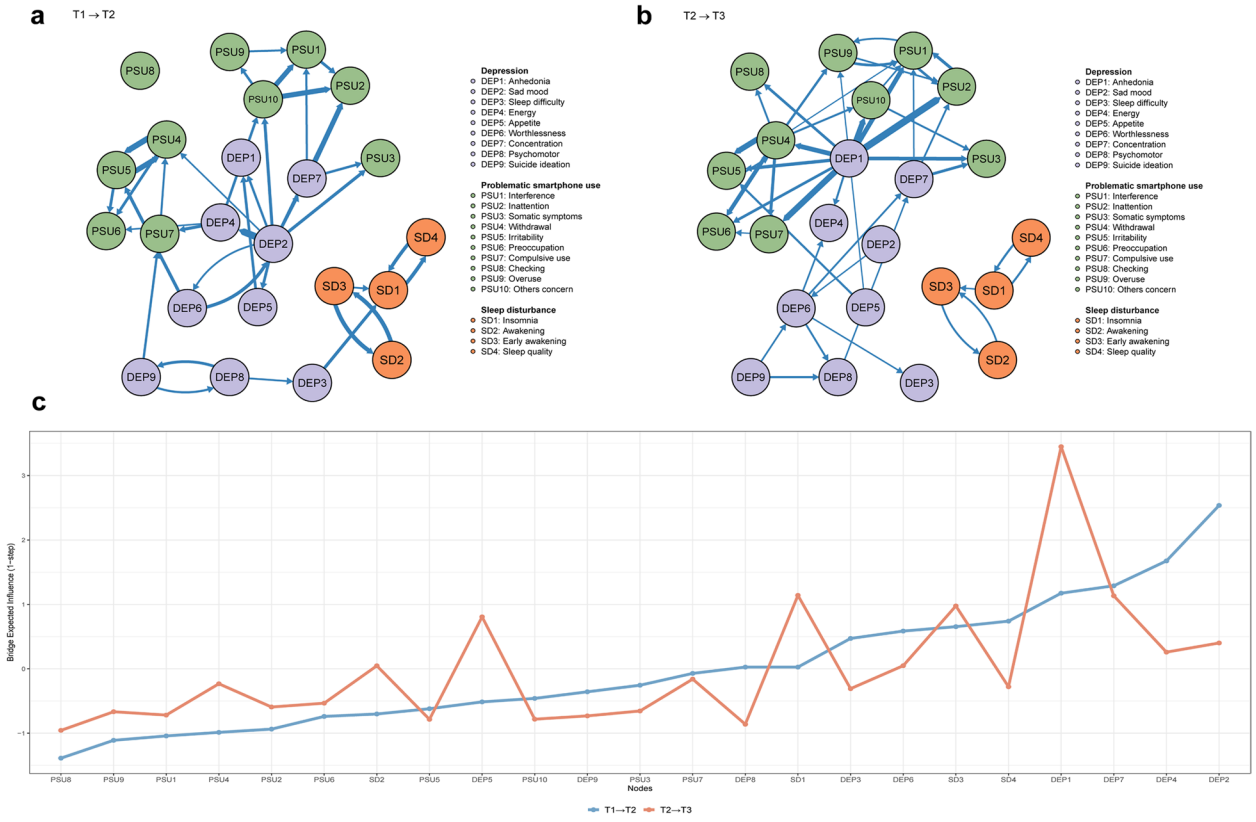


Fig. 3 CLPNs and bridge expected influence. Note. Panels **a**–**c** show the $T_1 \rightarrow T_2$ network structure (**a**), the $T_2 \rightarrow T_3$ network structure (**b**), and the bridge expected influence of the networks (**c**). Panels a and b illustrate the interrelationships among nodes across different time points, with fixed node layout to facilitate cross-time comparison. Arrows represent the lagged effects of one node on another (cross-lagged associations): blue indicates positive cross-lagged associations, and red indicates negative cross-lagged associations. Arrow thickness is proportional to the strength of the relationships. Autoregressive paths and covariates are not displayed. In addition, to enhance visual clarity, only paths with absolute standardized regression coefficients ≥ 0.07 are displayed. This threshold was used solely for visualization purposes and did not affect the statistical estimation or interpretation of the network. Complete edge-weight matrices, including all estimated paths, are provided in Supplementary Tables 3 and 4.

Discussion

Using a sample of Chinese college students, this study employed the RI-CLPM and the CLPN to examine the longitudinal associations among PSU, DEP, and SD. The RI-CLPM results indicated that, at the between-person level, the random intercepts of the three constructs were significantly correlated. At the within-person level, increases in depression consistently predicted subsequent increases in PSU and SD, whereas no significant reciprocal predictive relationship was found between PSU and SD. Further network analysis identified "Inattention," "Withdrawal," "Tolerance/Compulsive use," "Daily life affected," and "Early awakening" as key nodes with relatively high expected influence in the network, while "Concentration" emerged as a key bridge node across symptom domains.

The RI-CLPM results indicated that, at the between-person level, PSU, DEP, and SD were significantly and positively correlated, thereby supporting Hypothesis 1. This finding is consistent with previous evidence, suggesting that PSU, DEP, and SD generally exhibit stable covariation [67, 68]. However, at the within-person level, the RI-CLPM results only showed that DEP unidirectionally predicted subsequent PSU and SD, whereas the longitudinal mediating effect of DEP on the association between PSU and SD was not supported. The discrepancy between the between-person and within-person findings suggests that the association between PSU and SD may primarily reflect stable individual differences rather than dynamic within-person changes over time. The RI-CLPM distinguishes stable between-person differences from within-person fluctuations, indicating that associations observed at the between-person level do not necessarily imply corresponding dynamic processes at the within-person level [41, 69]. One possible explanation is that the between-person association reflects certain shared and relatively stable individual characteristics that simultaneously influence both PSU and SD. After these stable between-person differences are separated by the RI-CLPM, the remaining within-person fluctuations in PSU and SD may be insufficient to produce detectable cross-lagged effects over the six-month follow-up

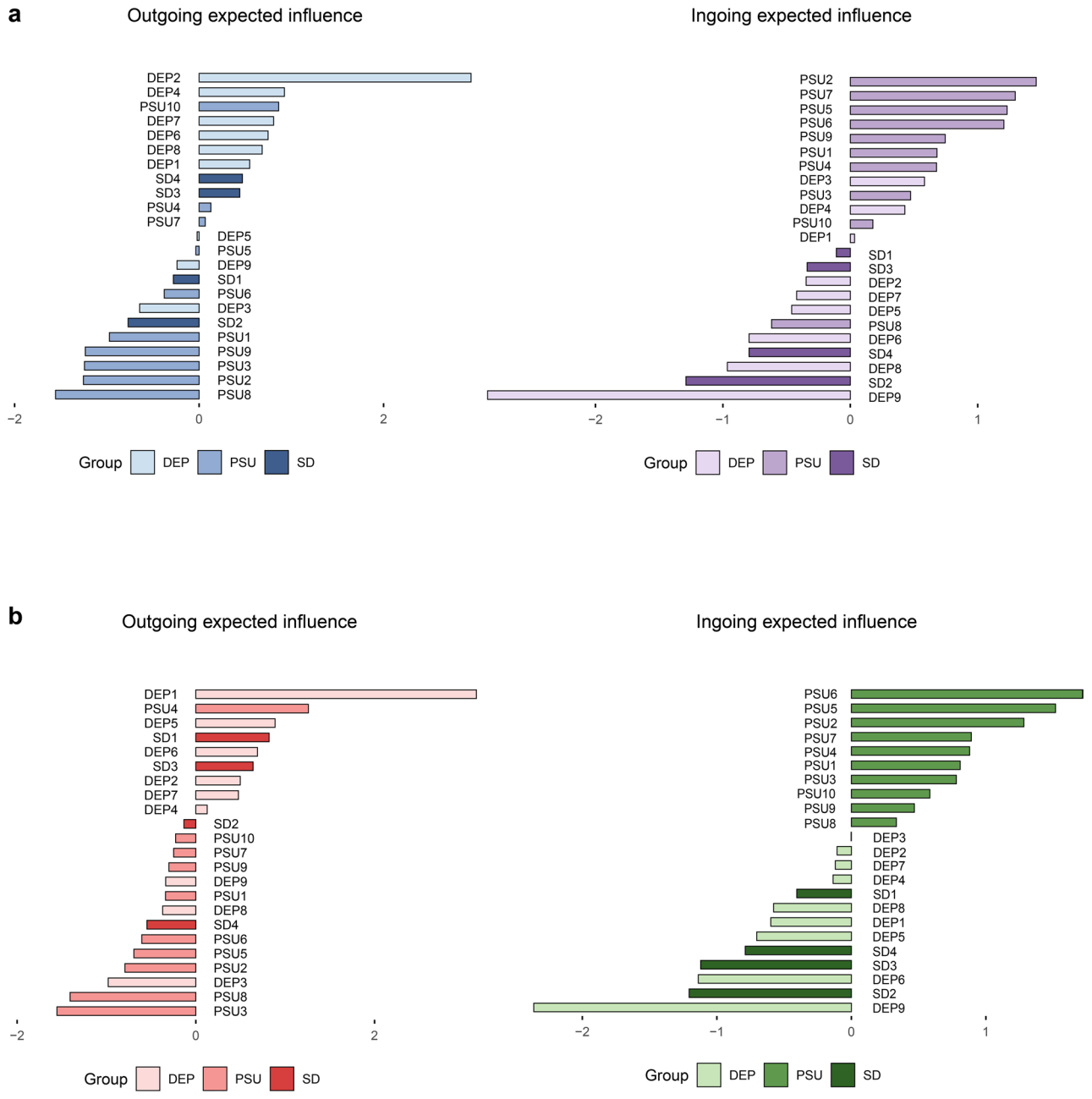


Fig. 4 Outgoing and ingoing expected influence of networks. Note. **a**, **b** Outgoing and ingoing expected influences of the $T_1 \rightarrow T_2$ network (**a**) and the $T_2 \rightarrow T_3$ network (**b**). These diagrams illustrate, across different periods, the nodes most likely to exert influence on others and those most susceptible to being influenced. Higher values indicate greater outgoing or ingoing expected influences of each node, respectively

period. It is worth noting that PSU showed relatively low explained variance in the present study (R^2 = 0.010 at T2; R^2 = 0.008 at T3), and its autoregressive path coefficients were also relatively small. RI-CLPM emphasizes capturing within-person temporal dynamics, in which autoregressive and cross-lagged effects jointly determine the evolution of deviations from an individual's expected level over time. According to the principle underlying the calculation of long-term effects, when autoregressive effects are weak, cross-lagged influences are less likely to accumulate across multiple time points, thereby limiting their long-term impact [70]. In addition to these methodological considerations, the present findings may also be related to the specific sociocultural and educational context of Chinese college students. Previous studies have shown that university students generally experience relatively high levels of academic stress [71], while mobile social media and smartphone use are highly prevalent among Chinese college students [72, 73]. These factors may be closely associated with individuals' smartphone

use behaviors and sleep patterns [72, 73]. Accordingly, contextual factors not directly included in the present model may also have influenced the within-person temporal dynamics between PSU and SD.

The CLPN analysis further identified several core nodes with relatively high expected influence within the integrated network, including PSU2 (Inattention), PSU4 (Withdrawal), PSU7 (Tolerance/Compulsive use), PSU10 (Daily life affected), and SD3 (Early awakening), as well as an important bridge node, DEP7 (Concentration). "Withdrawal", "Tolerance/Compulsive use", and "Daily life affected" reflect key addiction-related features of PSU [74, 75]. "Withdrawal" refers to withdrawal-like reactions that occur when individuals are unable to use their smartphones, such as anxiety, irritability, and emotional discomfort [76]. "Tolerance/Compulsive use" reflects diminished behavioral control, whereby individuals continue to use smartphones excessively despite being aware of potential negative consequences [77]. "Daily life affected" further indicates that problematic smartphone use has become externally observable and has begun to negatively affect real-life interpersonal interactions and daily functioning [78]. Compared with symptoms that merely reflect usage frequency or general smartphone use behaviors, these symptoms are more likely to interact persistently with emotional problems, sleep disturbance, and impairments in real-life functioning [76, 79], thereby playing a more important role in the dynamic associations among PSU, DEP, and SD. "Early awakening" often involves disruptions in sleep rhythm, persistent physiological and psychological arousal, and depression-related symptoms [80, 81]. It may therefore not only reflect sleep problems themselves but also be more likely to participate in the dynamic symptom interactions between DEP and PSU. In addition, PSU2 (Inattention) may be particularly sensitive to other symptoms, as smartphone use–related attentional difficulties are more likely to be jointly influenced by sleep problems, depressive mood, and other PSU symptoms [82–83]. Meanwhile, DEP7 (Concentration) emerged as a shared bridge node in both networks, suggesting that attentional/concentration difficulties may represent a key cognitive symptom linking PSU, DEP, and SD. Previous research has shown that PSU is associated with impaired attentional control and executive functioning [84], while sleep problems and depressive symptoms are also frequently accompanied by cognitive impairment [85, 86]. Accordingly, Concentration may be more likely to participate in the reciprocal influences among different symptom domains.

As a complement to the RI-CLPM findings, the CLPN analysis further highlighted several specific symptoms, including "Inattention", "Withdrawal", "Tolerance/Compulsive use", "Daily life affected", and "Early awakening", as persistent key nodes across different time points, while also identifying attentional/concentration difficulties represented by "Concentration" as an important bridge symptom linking PSU, DEP, and SD. It should be noted that the RI-CLPM and CLPN address different levels of research questions and therefore provide complementary perspectives. Specifically, the RI-CLPM examines longitudinal directional associations at the construct level by disentangling within-person and between-person variance, whereas the CLPN focuses on longitudinal interactions among individual symptoms and identifies those that play relatively central roles within the symptom network. Therefore, the absence of significant cross-lagged associations between PSU and SD at the construct level does not necessarily imply the absence of meaningful longitudinal connections at the symptom level. This dynamic perspective further suggests that addiction-related symptoms, sleep problems, and cognitive functioning may be closely intertwined over time, and that interventions targeting these key symptoms may provide useful implications for alleviating the co-occurrence of PSU, DEP, and SD.

Implications, limitations, and future directions

The present findings indicated that increases in DEP consistently predicted subsequent increases in PSU and SD. This suggests that future research on the relationship between PSU and SD should more fully consider the role of depressive symptoms and other emotional variables in both study design and data analysis. In addition, the present study further suggests that addiction-related PSU symptoms, sleep problems, and cognitive difficulties may engage in ongoing dynamic interactions over time.

The present findings also have implications for psychological counseling and intervention. Previous studies have shown that psychological interventions targeting depressive symptoms can improve individuals' emotional functioning [87, 88]. The present study further suggests that such interventions may also exert indirect effects on PSU and SD. Compared with interventions focusing solely on behavioral restriction or sleep management, interventions targeting emotion regulation, cognitive functioning, and addiction-related behavioral characteristics may be more effective in addressing the shared psychological and behavioral mechanisms underlying these problems. Specifically, the key nodes identified in the present study, including PSU-related symptoms such as Inattention, Withdrawal, Tolerance/Compulsive use, and Daily life affected, as well as the sleep-related symptom Early awakening, may represent important targets for intervention. Meanwhile, Concentration, identified as a bridge node, may also play an important role across different symptom domains. At the service-system level, the co-occurrence of PSU, DEP, and SD further suggests that universities and communities should establish

more integrated mental health service systems, including depressive symptom screening, sleep problem assessment, behavioral interventions, and stress-management support.

Several limitations should also be noted. First, all core variables were assessed using self-report questionnaires, which may be subject to social desirability and reporting biases. Future research could incorporate behavioral indicators, physiological measurements, or multi-informant assessments. Second, the sample included only university students and exhibited an imbalanced gender distribution, with female participants accounting for approximately 80% of the sample. Given that female college students generally report higher levels of depressive symptoms and sleep problems than male students, the predominance of female participants may also have influenced the overall magnitude of the observed associations. Therefore, future studies should replicate these findings in more diverse and gender-balanced samples. Third, although longitudinal measurement invariance of the PSU scale was supported based on the recommended criteria for changes in model fit indices, the relatively modest absolute fit indices of the PSU measurement model suggest that the current measurement model may not fully capture the covariance structure among PSU items. Therefore, findings involving PSU should be interpreted with appropriate caution. In addition, the overall explanatory power (R^2) of the RI-CLPM was relatively low, particularly for PSU, suggesting that within-person temporal dynamics may also be influenced by other unmeasured variables. Accordingly, the present findings should be interpreted with caution. Finally, although the present study employed a longitudinal design, the six-month interval between consecutive assessments may not have been optimal for capturing shorter-term fluctuations in PSU, DEP, and SD, which may vary in response to academic schedules, examination stress, or vacation-related lifestyle changes. Future studies should consider using shorter assessment intervals, more frequent measurement waves, or intensive longitudinal designs to better characterize the within-person temporal dynamics of these constructs.

Conclusion

Based on three-wave longitudinal data from Chinese college students, this study used RI-CLPM and CLPN to examine the longitudinal associations among PSU, DEP, and SD. At the between-person level, PSU, DEP, and SD were significantly and positively correlated. At the within-person level, increases in DEP consistently predicted subsequent increases in PSU and SD, whereas no stable bidirectional predictive relationship was found between PSU and SD. The CLPN analysis further identified several core nodes with relatively high expected influence, including "Inattention," "Withdrawal," "Tolerance/Compulsive use," "Daily life affected," and "Early awakening," while "Concentration" emerged as an important bridge node linking PSU, DEP, and SD. Overall, these findings suggest that depressive symptoms, sleep problems, cognitive difficulties, and addiction-related PSU symptoms may become intertwined during young adulthood development. Future research could incorporate longitudinal designs with more time points and further consider cross-cultural contexts and individual differences to examine the stability and generalizability of these dynamic associations. From a practical perspective, early identification and intervention targeting depressive symptoms, attentional/concentration difficulties, and core addiction-related PSU symptoms may help reduce the co-occurrence of PSU and SD, while also promoting adolescents' psychological well-being and sleep adjustment.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40359-026-05167-0.

Supplementary Material 1.

Informed consent

Informed consent was obtained from all participants prior to data collection.

Authors' contributions

Lujie Zhong: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization. Guoyi Zhu: Investigation, Methodology, Data curation, Validation, Writing – review & editing. Jiaying Li: Investigation, Methodology, Data curation, Project administration, Writing – review & editing. Xueqin Zhang: Investigation, Data curation, Writing – review & editing, Funding acquisition. Dongfang Wang: Supervision, Project administration, Writing – review & editing. Yun Li: Conceptualization, Supervision, Resources, Funding acquisition, Writing – review & editing.

Funding

This work was supported by the Young Scientists Fund of the National Natural Science Foundation of China (No. 32400894); the Basic and Applied Basic Research Foundation of Guangdong Province (No. 2023A1515110350); the Youth Project of the Guangdong Office of Philosophy and Social Science (No. GD24YXL06); the Guangzhou Basic and Applied Basic Research Program (Young Doctoral "Qihang" Project, Grant No. 2025A04J4058); the Characteristic New Think Tank of university in Guangdong Province: Guangdong-Hong Kong-Macao Greater Bay Area Medical and Health Industry High Quality Development Rule of Law Guarantee Research Center(No. 2024TSZK016); and "Sunflower Student Mental Health Promotion Action" jointly organized by the National Center for Mental Health, China, China Education Development Foundation, and the Center for Student Services and Development, Ministry of Education, P.R. China. (GX25A003). The funders had no role in study design, data collection, analysis, or the decision to submit the manuscript.

Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was reviewed and approved by the Ethics Committee of the School of Psychology, South China Normal University (Approval No.: SCNU-PSY-2023-345). All procedures were performed in accordance with the Declaration of Helsinki and relevant institutional guidelines.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

Received: 9 June 2026 / Accepted: 7 July 2026
Published online: 22 July 2026

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